

HackTheBox Academy

Web Requests Module

Course name: Certified Junior Cybersecurity Associate

POST REQUEST

POST request is a request that web applications utilize when they need to move the user parameters or transfer files from the URL.

The task that we have for this section is: Obtain a session cookie through a valid login, and then use the cookie with cURL to search for the flag through a JSON POST request to '/search.php'

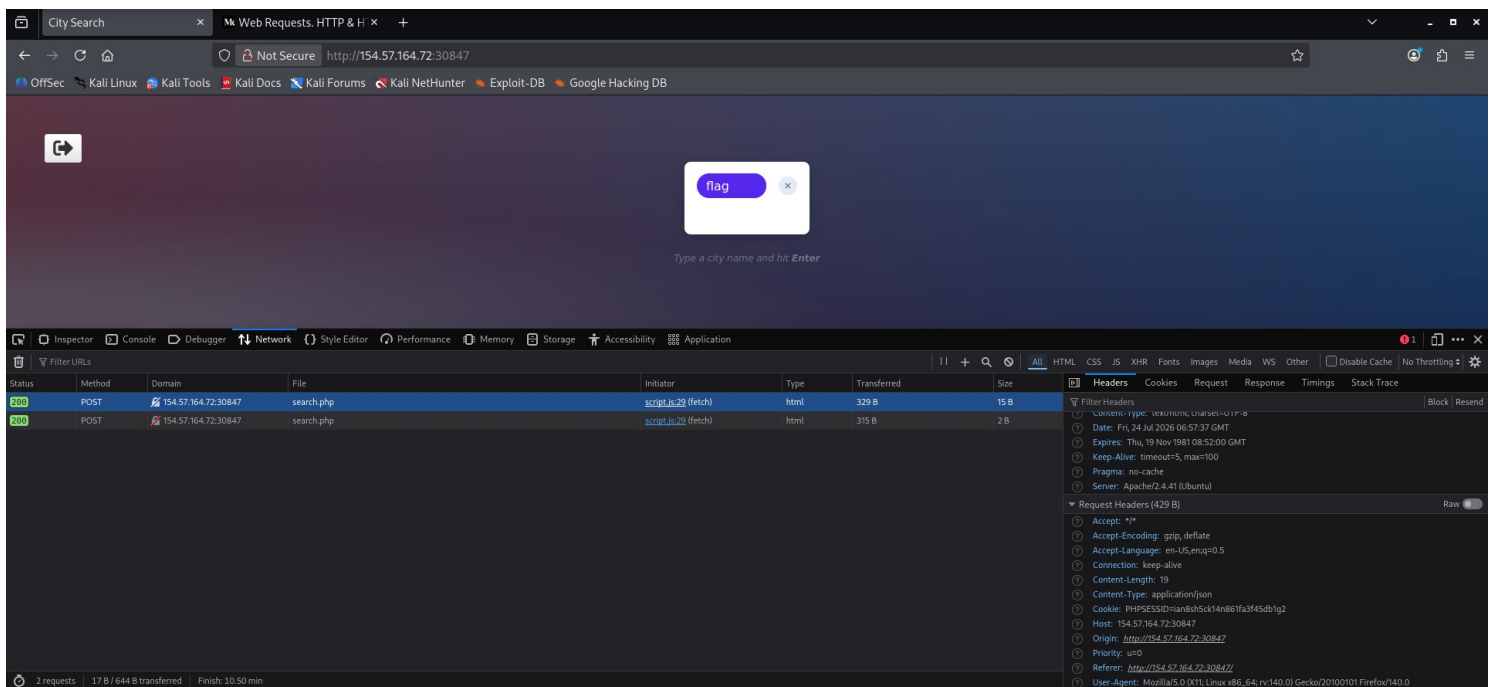
We will first start by spawning the HackTheBox target IP and Port address: 154.57.164.72:30847

In the browser we will type the IP and Port into our browser to access the city search:

<http://154.57.164.72:30847>

We will log in using the credentials provided: admin:admin.

To capture the cookie we can obtain it on the Network tab in DevTools.



```
Cookie: PHPSESSID=ian8sh5ck14n861fa3f45db1g2
Host: 154.57.164.72:30847
Origin: http://154.57.164.72:30847
```

As we can see in the image, the cookie is: PHPSESSID=ian=8sh5ck14n861fa3f45db1g2

The next step is to type the command: `curl -X POST -d '{"search":"'flag'"}' -b`

```
Session Actions Edit View Help
$ curl -X POST -d '{"search":"flag"}' -b 'PHPSESSID=ian8sh5ck14n861fa3f45db1g2' -H 'Content-Type: application/json' http://154.57.164.72:30847/

<!DOCTYPE html>
<html Lang=en>

<head>
  <meta charset="UTF-8">
  <title>City Search</title>
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/meyer-reset/2.0/reset.min.css">
  <link rel="stylesheet" href="https://fonts.googleapis.com/css?family=Roboto:400,500,700">
  <link rel="stylesheet" href="https://static.fontawesome.com/css/fontawesome-app.css">
  <link rel="stylesheet" href="https://pro.fontawesome.com/releases/v5.2.0/css/all.css">
  <link rel="stylesheet" href="/style.css">
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/normalize/5.0.0/normalize.min.css">
  <script src="https://cdnjs.cloudflare.com/ajax/libs/prefixfree/1.0.7/prefixfree.min.js"></script>
</head>

<body>
  <div class="icon" style="position: absolute; top: 50px; left: 50px;">
    <a href="/?logout" class="btn btn-success"><i class="fab fas fa-sign-out-alt fa-2x"></i></a>
  </div>
  <div>
    <div class="search">
      <div class="bar">
        <div class="icon">
          <i></i>
        </div>
        <input type="text">
      </div>
      <div class="close"></div>
      <ul id="results_list">
      </ul>
    </div>
    <em>Type a city name and hit <strong>Enter</strong></em>
  </div>
  <script src="https://cdnjs.cloudflare.com/ajax/libs/jquery/3.3.1/jquery.min.js"></script>
  <script src="/script.js"></script>
</body>
</html>

$ curl -X POST -d '{"search":"flag"}' -b 'PHPSESSID=ian8sh5ck14n861fa3f45db1g2' -H 'Content-Type: application/json' http://154.57.164.72:30847/search.php
{"flag": "HTB[p08t_r3p3kt3r]"}
$ curl -X POST -d '{"search":"flag"}' -b 'PHPSESSID=ian8sh5ck14n861fa3f45db1g2' -H 'Content-Type: application/json' http://154.57.164.72:30847/
```

'PHPSESSID=ian=8sh5ck14n861fa3f45db1g2' -H 'Content-Type: application/json'
<http://154.57.164.72:30847/search.php>

once we execute the command we get the flag.

Command breakdown:

- curl - command line tool and library that primarily supports HTTP along with other protocols. It is used to transfer data to and from servers using various network protocols.
- -X POST – flag, that is used to send the request.
- -d '{"search":"flag"}' – sends data in the request body, here JSON object contains the search value "flag".
- 'PHPSESSID=ian=8sh5ck14n861fa3f45db1g2' – session cookie.
- -H 'Content-Type: application/json' – adds header telling the server that the request body is JSON format.
- <http://154.57.164.72:30847/search.php> – endpoint receiving and processing the request.